

Which training videos are worth keeping?

EgoScore · EgoVerse Data Optimization & Evaluation Suite, Track 1 · github.com/sahilmahendrakar/egoscore

Both sides picked the same 130 videos out of 572

THE PROBLEM

EgoVerse is a library of videos of people doing everyday tasks, filmed from a camera on their head. Robots learn from them. It holds hundreds of thousands of clips, and more is not automatically better: some are broken, and many are near-copies of each other.

WHAT WE BUILT

1 Throw out the broken ones.

Tracking glitches, hands off to the side, or someone who left the camera running for ten minutes.

2 Pick a varied quarter of the rest.

Instead of picking at random, choose clips unlike each other — different people, rooms and ways of moving.

3 Check whether it worked.

Train the same small model on each selection and see which better predicts what a person does next.



Each dot is one video. Dots that sit close together are videos of someone moving in a similar way.

THE RESULT

The varied quarter beat the random quarter in all 40 of our 40 comparisons, cutting the model's prediction error by 3.5%.

How do we know it's real?

A 4% difference is small enough to be luck, so we also ran a selection built to be bad on purpose: nearly every clip from a handful of people and rooms.

It lost 39 of 40 times, doing +6% worse. So our test can tell a good selection from a bad one, and the smaller result is a real effect rather than noise.

What we're not claiming

Using all the data is still best. Picking well does not beat having more.

What it does do: at a quarter of the data, a varied pick closes about 45% of the gap between a random quarter and using everything. That is the number that matters when you can't have everything.

What we found in the data

Some labs film one task per clip. Others leave the camera running for a whole session and call it one clip. A clip is 90 seconds in one lab and 11 in another — 13.9% of the biggest collection is really a whole session.

Cutting those makes the model 5% better, using the same amount of footage.

Where we'd push back

We measure how well a model predicts human hand movement, not whether a robot succeeds at the task. It is a stand-in — and the people who built EgoVerse use the same one, for the same reason.

We tested one task in one lab. We can't promise this carries over to others.